

Professional Online Academy Vu Topper RM



CS502-Fundamentals Of Algorithms Update MCQ'S Final Term



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Question No:1

(Marks:1)

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Which of the following problems is NOT solved using dynamic programming?

- A. Matrix chain multiplication problem
- B. Edit distance problem
- C. Fractional knapsack problem**
- D. 0/1 knapsack problem

Question No:2

(Marks:1)

Vu-Topper RM

Breadth-first search is a traversal strategy for _____.

- A. Tree
- B. Link List
- C. Array
- D. Graph**

Question No:3

(Marks:1)

Vu-Topper RM

In a digraph, the number of edges coming in of a vertex is called the _____ of that vertex.

- A. Node
- B. In-degree**
- C. Out-degree
- D. Complete degree

Question No:4

(Marks:1)

Vu-Topper RM

In an undirected graph, we say that an edge is _____ on a vertex if the vertex is an endpoint of the edge.

- A. Tree
- B. Directed
- C. Undirected
- D. Incident**

Question No:5

(Marks:1)

Vu-Topper RM

Which graph traversal algorithm uses a stack to keep track of vertices?

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A. Depth First Search

- B. Greedy Algorithm
- C. Divide and Conquer
- D. Breadth First Search

Question No:6

(Marks:1)

Vu-Topper RM

In general, a graph $G = (V, E)$ consists of a _____ and E , a binary relation on V called edges.

A. Finite set of vertices V

- B. Infinite set of objects
- C. Infinite set of nodes
- D. Infinite set of vertices V

Question No:7

(Marks:1)

Vu-Topper RM

Each time we traverse graph by Breadth-first search algorithm, we count the distance from _____

- A. left most node
- B. neighbors of the starting node
- C. starting node**
- D. right most node

Question No:8

(Marks:1)

Vu-Topper RM

For traversing the graphs, _____ can be visualized as a wave front propagating inward towards root node.

- A. Binary search
- B. Linear search
- C. Breadth-first search**
- D. Depth-first Search

Question No:9

(Marks:1)

Vu-Topper RM

In a digraph, the number of edges coming out of a vertex is called the _____ of that vertex.

- A. Node

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- B. In-degree
- C. Out-degree**
- D. Complete degree

Question No:10 (Marks:1) **Vu-Topper RM**
Brute-force is a simple strategy for computing _____.

- A. shortest path**
- B. Longest path
- C. Complete path
- D. Easiest path

Question No:11 (Marks:1) **Vu-Topper RM**
In general in comparison with Fractional Knapsack problem,

- a. knapsack problem is very easy to solve
- b. knapsack problem is hard to solve**
- c. We cannot compare them
- d. Both are easy to solve

Question No:12 (Marks:1) **Vu-Topper RM**
_____ is a good way to model some sort of “connection” or “relationship” or “interaction” between pairs of objects taken from a set of objects.

- A. Vertices
- B. Tree
- C. Graph**
- D. Node

Question No:13 (Marks:1) **Vu-Topper RM**
For a digraph $G = (V, E)$, Sum of in-degree(v) _____

- A. $<$ Sum of out-degree(v)
- B. = Sum of out-degree(v)**
- C. Sum of out-degree(v)

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D. Not equal to Sum of out-degree(v)

Question No:14

(Marks:1)

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Graphs are important _____ model for many application problems.

- A. Unpredictable
- B. Mathematical**
- C. Unsystematic
- D. Haphazard

Question No:15

(Marks:1)

Vu-Topper RM

Which of the following algorithms solves the fractional knapsack problem most effectively?

- A. Backtracking
- B. Greedy algorithm**
- C. Divide and conquer
- D. Dynamic programming

Question No:16

(Marks:1)

Vu-Topper RM

A _____ cycle is a cycle that visits every edge of the graph exactly once.

- A. Undirected
- B. Loop
- C. Eulerian**
- D. Directed

Question No:17

(Marks:1)

Vu-Topper RM

In Fractional Knapsack problem, one is allowed to take fraction of an item for _____

- A. None of the given options
- B. for fraction of the value
- C. Both, fraction of the weight and value
- D. for fraction of the weight**

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Question No:18

(Marks:1)

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In Fractional Knapsack problem, the goal is to _____

- A. minimize the value of items without exceeding the total weight limit of W
- B. maximize the value of items without exceeding the total weight limit of W**
- C. maximize the value of items even if exceeding the total weight limit of W
- D. minimize the value of items even if exceeding the total weight limit of W

Question No:19

(Marks:1)

Vu-Topper RM

Traversing a graph means visiting _____ in the graph.

- A. every node**
- B. more than one node
- C. no node
- D. at least o

Question No:20

(Marks:1)

Vu-Topper RM

There are _____ ways of representing graphs.

- A. 1
- B. 3
- C. 2
- D. 4**

Question No:21

(Marks:1)

Vu-Topper RM

Which of the following is true about graph?

- A. A graph may contain no vertices and many edges
- B. A graph may contain many edges and no vertices
- C. A graph may contain no edges and many vertices**
- D. A graph may contain no edges and no vertices

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Question No:22

(Marks:1)

Vu-Topper RM

Each vertex can reach every other vertex is called _____.

- A. Unconnected graph
- B. Unordered graph
- C. Ordered graph
- D. Connected graph**

Question No:23

(Marks:1)

Vu-Topper RM

In chain matrix multiplication, the order of the matrices _____.

- A. is equal
- B. is reverse**
- C. can not be changed
- D. can be changed

Question No:24

(Marks:1)

Vu-Topper RM

In Knapsack Problem, each item must be entirely accepted or rejected, is called _____ problem.

- A. Linear
- B. Fractional
- C. 0-1**
- D. Optimal

Question No:25

(Marks:1)

Vu-Topper RM

In Dynamic Programming based solution of Knapsack Problem, if we decide to take an object 'i', then we gain _____.

- A. W(Total Weight of Knapsack)
- B. V (Total Value of all items)
- C. v_i (Value of object i)**
- D. None of the given option

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Question No:26

(Marks:1)

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In chain matrix multiplication, table is filled _____ to find the multiplication of matrix.

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- A. row wise
- B. column wise
- C. diagonally

D. bottom-to-up **Page 86**

Question No:27 **(Marks:1)** **Vu-Topper RM**

In chain matrix multiplication, if there are n items, there are _____ ways in which outer most pair of parentheses can placed.

- A. n^2
- B. $2n$
- C. $n+1$

D. $n-1$ **Page 85**

Question No:28 **(Marks:1)** **Vu-Topper RM**

Catalan numbers are related the number of different _____ on ' n ' nodes.

- A. Arrays
- B. linked lists

C. binary trees **Page 85**
D. functions

Question No:29 **(Marks:1)** **Vu-Topper RM**

What is the value of k in the given equation?

$$m[1, 5] = m[1, 3] + m[4, 5] + p_0 \cdot p_3 \cdot p_5$$

- A. 3** **ok**
- B. 1
- C. 4
- D. 5

Question No:30 **(Marks:1)** **Vu-Topper RM**

The running time of brute-force algorithm to solve Knapsack problem is _____.

- A. $O(n!)$
- B. $O(2^n)$**

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- C. $O(n)$
- D. $O(n^3)$

Question No:31

(Marks:1)

Vu-Topper RM

What is the value of k in the given equation?

$$m[2, 5] = m[2, 3] + m[4, 5] + p_1 \cdot p_3 \cdot p_5$$

- A. 3
- B. 5
- C. 2**
- D. 4

Question No:32

(Marks:1)

Vu-Topper RM

We can't utilize a divide-and-conquer strategy for Chain Matrix Multiplication because _____.

- A. Size of data is not given
- B. We can easily perform it in linear time
- C. We use divide and conquer for sorting only
- D. We do not know the optimum k**

Question No:33

(Marks:1)

Vu-Topper RM

Suppose we have 5 matrices A, B, C, D and E. What is the correct expansion of $m[3,4]$ in chain matrix multiplication?

- A. $m[3, 4] = m[3, 4] + m[4, 4] + p_2 \cdot p_3 \cdot p_4$
- B. $m[3, 4] = m[3, 3] + m[3, 4] + p_2 \cdot p_3 \cdot p_4$
- C. $m[3, 4] = m[3, 3] + m[4, 4] + p_0 \cdot p_3 \cdot p_4$
- D. $m[3, 4] = m[3, 3] + m[4, 4] + p_2 \cdot p_3 \cdot p_4$**

Question No:34

(Marks:1)

Vu-Topper RM

Time complexity of activity selection algorithm is _____.

- A. $O(\log N^2)$
- B. $O(N \log N)$**
- C. $O(\log N)$
- D. $O(N)$

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Question No:35

(Marks:1)

Vu-Topper RM

If we have 6 metrics in chain matrix multiplication problem then the number of table entries must be?

- A. 12
- B. 25
- C. 30

D. 36 **Google**

Question No:36

(Marks:1)

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Counting Money problem:

- A. Can be solved by simple recursive algorithm
- B. Can be optimally solved by greedy algorithm**
- C. can not be optimally solved by greedy algorithm
- D. Can not be solved by brute force algorithm

Question No:37

(Marks:1)

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Time complexity of Dynamic Programming based algorithm for computing the minimum cost of Chain Matrix Multiplication is

_____.

- A. Log n
- B. n
- C. n^2 (n square)

D. n^3 (ncube) **Page 90**

Question No:38

(Marks:1)

Vu-Topper RM

_____ items are not allowed in the 0/1 knapsack.

- A. Lighter
- B. Whole
- C. Weighty
- D. Fractional**

Question No:39

(Marks:1)

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If matrix A of dimension $p \times q$ is multiplied with matrix B of dimension $q \times r$, then each entry in resultant matrix takes _____ time.

A. O (q)

Page 84

B. (1)

C. ($p \times q$)

D. ($q \times r$)

Question No:40

(Marks:1)

Vu-Topper RM

In Activity scheduling algorithm, the width of a rectangle _____

A. Directs towards recursion

B. Indicates the duration of an activity **ok**

C. Is always ignored

D. Should be maximized

Question No:41

(Marks:1)

Vu-Topper RM

What will be value of the matrices product $(A A)A$? if $A = 5 \times 6$ $A = 6 \times 7$
 $A = 7 \times 2$.

A. 290

B. 280

C. 250

D. 270

Question No:42

(Marks:1)

Vu-Topper RM

Which of the following algorithms is the best approach for solving Huffman codes?

A. Divide/conquer algorithm

B. Brute force algorithm

C. Greedy algorithm

D. Exhaustive search

Question No:43

(Marks:1)

Vu-Topper RM

Using base condition we set all $m[i,i] =$ _____ ?

A. 1

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B. 0

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C. ∞

D. -1

Question No:44

(Marks:1)

Vu-Topper RM

What will be value of the matrices product $(A A)A$? if $A = 5 \times 3$ $A = 3 \times 7$
 $A = 7 \times 2$.

A. 180

B. 175

C. 88

D. 80

Question No:45

(Marks:1)

Vu-Topper RM

In Huffman encoding fixed-length codes are popular because it is very easy to break up a _____ into its individual _____.

A. Number, String

B. Floating number, String

C. Character, Number

D. String, Characters

Question No:46

(Marks:1)

Vu-Topper RM

The _____ is a problem for which the greedy algorithm approach provides an optimal solution.

A. Knapsack Problem

B. NP complete problem

C. Dynamic programming

D. Activity selection

Question No:47

(Marks:1)

Vu-Topper RM

Consider the string "abcdaacac". if the string is coded with ASCII codes, the message length would be _____.

A. 72 bits

B. 60 bits

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- C. 90 bits
- D. 70 bits

Question No:48

(Marks:1)

Vu-Topper RM

The formula for calculating the Catalan number is _____.

☐	$C(n) = \frac{1}{n+1} \binom{n}{2n}$	
☐	$C(n) = \frac{1}{n-1} \binom{n}{2n}$	
☒	$C(n) = \frac{1}{n+1} \binom{2n}{n}$	Page 85
☐	$C(n) = \frac{1}{n-1} \binom{2n}{n}$	

Question No:49

(Marks:1)

Vu-Topper RM

Dynamic programming formulation of the matrix chain multiplication problem will store the solutions of each sub problem in a(n):

- A. Class
- B. Array
- C. Table**
- D. Variable

Question No:50

(Marks:1)

Vu-Topper RM

Consider two matrices A and B, where A is 10*30 and B is 30*40. What is correct number of multiplications require to multiply A and B?

- A. 30*40
- B. 10*30
- C. 30*30
- D. 10*30*40**

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Question No:51

(Marks:1)

Vu-Topper RM

Which of the following is true about Huffman Coding?

- A. In Huffman coding, no code is prefix of any other code**
- B. Huffman Codes may not be optimal lossless codes in some cases
- C. Huffman coding is not an efficient coding way
- D. Huffman coding may become lossy in some cases

Question No:52

(Marks:1)

Vu-Topper RM

The optimal solution to a problem is a combination of optimal solutions to its subproblems. This is known as _____.

- A. Principle of feasibility
- B. Principle of locality
- C. Principle of computation
- D. Principle of optimality**

Question No:53

(Marks:1)

Vu-Topper RM

You are given a knapsack that can carry a maximum weight of 60. There are 4 items with weights {20, 30, 40, 70} and values {90, 80, 100, 250}. What is the maximum value of the items you can carry using the knapsack?

- A. 270**
- B. 190
- C. 520
- D. 170

Question No:54

(Marks:1)

Vu-Topper RM

In Huffman Encoding, the characters with smallest probabilities are placed at the _____ depth of the tree.

- A. Root
- B. Minimum**
- C. Maximum
- D. Average

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Question No:55

(Marks:1)

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Which one of the following problems can be solved using dynamic problem?

- A. Bubble sort problem
- B. Greedy search problem
- C. Fractional knapsack problem
- D. Matrix chain multiplication problem**

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Question No:56

(Marks:1)

Vu-Topper RM

Huffman algorithm uses a _____ to generate a prefix code T that minimizes the expected length $B(T)$ of the encoded string.

- A. Heuristic Approach
- B. Null Approach
- C. Greedy Approach**
- D. Dynamic Programming

Question No:57

(Marks:1)

Vu-Topper RM

If a matrix has three rows and two columns, then dimensions of matrix will be:

- A. 3×2**
- B. 2×3
- C. 3×3
- D. 2×2

Question No:58

(Marks:1)

Vu-Topper RM

In general, the activity selection problem is to select a _____.

- A. maximum-size set of mutually non-interfering activities**
- B. minimum-size set of mutually non-interfering activities
- C. maximum-size set of interfering activities
- D. minimum-size set of interfering activities

Question No:59

(Marks:1)

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We can multiply two matrices A and B only when they are compatible which means:

- A. Number of rows and columns do not matter
- B. Number of rows in A must be equal to number of rows in B
- C. Number of columns in A must be equal to number of rows in B**
- Page 84
- D. Number of columns in A must be equal to number of columns in B

Question No:60 (Marks:1) **Vu-Topper RM**

In context of activity selection algorithm, time is dominated by sorting of the activities by _____.

- A. Finish Times**
- B. Start Times
- C. Average Times
- D. CPU Burst Times

Question No:61 (Marks:1) **Vu-Topper RM**

Algorithm allows negative weights edges and no negative cost cycles.

- A. Brute-force technique
- B. Bellman-Ford** Page 159
- C. Dijkstra's
- D. Print

Question No:62 (Marks:1) **Vu-Topper RM**

In Prim's algorithm, we start with the _____ vertex r; it can be any vertex.

- A. Root** Page 149
- B. Pivot
- C. Leaf
- D. Negative

Question No:63 (Marks:1) **Vu-Topper RM**

Overall Running time of Prim's algorithm is _____.

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- A. $\Theta(E \log E)$
- B. $\Theta(E \log V)$
- C. $\Theta((V+E) \log V)$**
- D. $\Theta((V+E) \log E)$

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Question No:64

(Marks:1)

Vu-Topper RM

Problems such as the shortest route between cities can be solved efficiently by modeling the road map as a _____.

- A. Tree
- B. Stack
- C. Graph**
- D. Linked list

Google

Question No:65

(Marks:1)

Vu-Topper RM

In Kruskal's algorithm, the next edge is added to viable set A, if its adding does not induce a/an _____.

- A. Tree
- B. Edge
- C. Cycle**
- D. Vertex

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Question No:66

(Marks:1)

Vu-Topper RM

Bellman-Ford algorithm is used to solve _____ problems.

- A. Flow of networking
- B. All pair shortest path
- C. Single source shortest path**
- D. Double source shortest path

Google

Question No:67

(Marks:1)

Vu-Topper RM

In Prim's algorithm, If the color of a vertex is _____, then it is in S otherwise not.

- A. Blue
- B. Gray

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Question No:68**(Marks:1)****Vu-Topper RM**

Keeping in mind the shortest-path, if given scenarios occur in computer networks like the internet where data packets have to be routed. The vertices are _____ and Edges are _____ which may be wired or wireless.

- A. Internet, routers
- B. Routers, internet
- C. Communication links, routers

D. Routers, communication links**Page 153****Question No:69****(Marks:1)****Vu-Topper RM**

Dijkstra's algorithm :

- A. Has greedy approach to find all shortest paths
- B. Has both greedy and Dynamic approach to find all shortest paths
- C. Has greedy approach to compute single source shortest paths to all other vertices**
- D. Has both greedy and dynamic approach to compute single source shortest paths to all other vertices.

Page 154**Question No:70****(Marks:1)****Vu-Topper RM**

In Dijkstra's algorithm, initially the estimated value from source vertex to any vertex v is:

- A. One (1)
- B. Zero (0)
- C. Infinity (∞)**
- D. Minus one (-1)

Page 154**Question No:71****(Marks:1)****Vu-Topper RM**

From the given option's which one is correct regarding the time complexity of Dijkstra's algorithm:

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A. $O(N)$

B. $O(N^2)$

Google

C. $O(N^3)$

D. $O(\log N)$

Question No:72

(Marks:1)

Vu-Topper RM

In Bellman-Ford Algorithm, relaxation applies to _____ of the graph.

A. Every edge

Page 159

B. Every Vertices

C. Only First edge

D. Only First Vertices

Question No:73

(Marks:1)

Vu-Topper RM

The key[u] is the weight of the _____ going from u to any vertex in S.

A. Edge

B. lighter edge

C. lightest edge

Page 151

D. heavier edge

Question No:74

(Marks:1)

Vu-Topper RM

In Bellman-Ford Algorithm, path consists of at most _____ edges.

A. $V-1$

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B. $E+1$

C. $V+1$

D. $E-1$

Question No:75

(Marks:1)

Vu-Topper RM

Overall time for Kruskal algorithm is:

A. $\Theta(\log E)$

B. $\Theta(V \log E)$

C. $\Theta(E \log V)$

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D. $\Theta(\text{ElogE})$

Question No:76

(Marks:1)

Vu-Topper RM

A graph may contain -----.

A. No MST

B. One or zero MST

C. Exactly one MST

D. More than one MST

Google

Question No:77

(Marks:1)

Vu-Topper RM

In which algorithm, information of shortest path is propagated sequentially along each shortest path in the graph.

A. Prim's

B. Dijkstra's

C. Bellman-Ford

Page 16

D. Brute-force technique

Question No:78

(Marks:1)

Vu-Topper RM

Edge weights can be interpreted as distance _____.

A. in breadth-First Search

B. in Queue's

C. in the shortest-paths

Page 153

D. in depth-First Search

Question No:79

(Marks:1)

Vu-Topper RM

In the shortest-paths problem, we are given a weighted of _____ $G = (V, E)$.

A. Line graph

B. Directed graph

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C. Weighted graph

D. Un-directed graph

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Question No:80

(Marks:1)

Vu-Topper RM

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the _____ has relatively few

_____.

A. tree, edges

B. graph, edges

Google

C. tree, branches

D. graph, branches

Question No:81

(Marks:1)

Vu-Topper RM

From given algorithms which one considered as best for finding the shortest-path:

A. BFS

B. DFS

C. Dijkstra's algorithm

Page 154

D. Bellman-Ford algorithm

Question No:82

(Marks:1)

Vu-Topper RM

Finding the faster result of the shortest path from u to v for every pair of vertices u and v we use _____.

A. both I and II

B. All-pairs shortest-paths problem

Page 153

C. Two-pairs shortest-paths problem

D. Single-pairs shortest-paths problem

Question No:83

(Marks:1)

Vu-Topper RM

The running time of Bellman-Ford Algorithm is _____.

A. $\Theta(V^2)$

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B. $\Theta(E + E)$

C. $\Theta(V + E)$

D. $\Theta(V + V)$

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Question No:84

(Marks:1)

Vu-Topper RM

Dijkstra's algorithm is a simple _____ algorithm for computing the single-source shortest-paths to all other vertices.

A. Greedy

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B. Brute-Force

C. Bellman-Ford

D. Divide and conquer

Question No:85

(Marks:1)

Vu-Topper RM

In Prim's algorithm, we will make use of _____.

A. List

B. Array

C. Stack

D. Priority Queue

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Question No:86

(Marks:1)

Vu-Topper RM

What is the time complexity to extract a vertex from the priority queue in Prim's algorithm?

A. $(\log E)$

B. (V)

C. $(V+E)$

D. $O(\log V)$

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Question No:87

(Marks:1)

Vu-Topper RM

Bellman-Ford Algorithm does not allow $G(\text{graph})$ to have _____.

A. positive cost cycles

B. negative cost cycles

Page 159

C. positive weights edges

D. negative weights edges

Question No:88

(Marks:1)

Vu-Topper RM

Which of the following statement is false about Dijkstra's Algorithm?

A. It works on a weighted directed graph

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B. It is used to solve Single-source shortest path

C. It can be applied on graphs having a negative weight function

D. Its implementation in data structure is possible through the priority queue

Question No:89

(Marks:1)

Vu-Topper RM

In _____ algorithm(s), at any time, the subset of edges A forms a single tree.

A. BFS

B. Prim's

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C. Kruskal's

D. kruskal's and Prim's

Question No:90

(Marks:1)

Vu-Topper RM

Floyd-Warshall Algorithm is based on _____.

A. Complexity theory

B. Greedy Approach

C. Divide and Conquer

D. Dynamic Programming

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Question No:91

(Marks:1)

Vu-Topper RM

Dijkstra's Algorithm cannot be applied on _____.

A. unweighted graphs

B. directed and weighted graphs

C. undirected and unweighted graphs

D. graphs having negative weight function

Question No:92

(Marks:1)

Vu-Topper RM

Bellman-Ford algorithm is slower than _____.

A. Prim's

B. Dijkstra's

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C. Graph Algorithm

D. Brute-force technique

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Question No:93

(Marks:1)

Vu-Topper RM

In Bellman-Ford Algorithm, relaxation applies to every edge of the graph and repeat this _____ time.

A. V-1

Page 159

B. E-1

C. V+1

D. E+1

Question No:94

(Marks:1)

Vu-Topper RM

Which of the following is used in the data structure for implementing Dijkstra's Algorithm?

A. Stack's

B. Max heap

C. Priority queue

Google

D. Circular queue

Question No:95

(Marks:1)

Vu-Topper RM

In Prim's algorithm, If there is no edge from u to a vertex in S, we set the key value to _____.

A. 0

B. -1

C. ∞

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D. 1

Question No:96

(Marks:1)

Vu-Topper RM

Prim's algorithm is based on ----- strategy.

A. Greedy

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B. Exponential

C. Divide and Conquer

D. Dynamic programming

Question No:97

(Marks:1)

Vu-Topper RM

Dijkstra's Algorithm is used to solve _____ problems.

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- A. Sorting & searching
- B. All-pair shortest path
- C. Multi-source shortest path

D. Single-source shortest path

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Question No:98

(Marks:1)

Vu-Topper RM

The process of updating estimates in Dijkstra's algorithm is called _____.

- A. Insertion
- B. Updating
- C. Relaxation**
- D. Amendment

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Question No:99

(Marks:1)

Vu-Topper RM

_____ is commonly the running time of Dijkstra's Algorithm using the binary heap method.

- A. $\Theta(\log E)$
- B. $\Theta(V \log)$
- C. $\Theta(E \log V)$**
- D. $\Theta(\log V)$

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Question No:100

(Marks:1)

Vu-Topper RM

Kruskal's algorithm works by adding _____ in increasing order of weight (lightest edge first).

- A. Trees
- B. Edges**
- C. Vertices
- D. Weights

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Question No:101

(Marks:1)

Vu-Topper RM

Dijkstra's algorithm works on a weighted directed graph $G = (V, E)$ in which all _____ weights are non-negative.

- A. links

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B. Edges**Page 154**

C. Nodes

D. Vertices

Question No:102**(Marks:1)****Vu-Topper RM**

In Kruskal's algorithm, the next _____ is not added to viable set A, if its adding induce a/an cycle.

A. Tree

B. Edge

C. Cycle**Page 147**

D. Vertex

Question No:103**(Marks:1)****Vu-Topper RM**

The breadth-first-search algorithm is a shortest-path algorithm that works on _____ graphs.

A. Directed

B. Weighted

C. Un-directed

D. Un-weighted**Page 153****Question No:104****(Marks:1)****Vu-Topper RM**

An un-weighted graph can be considered as a graph in which every edge has weight _____ unit.

A. 1**Page 153**

B. 3

C. 5

D. 7

Question No:105**(Marks:1)****Vu-Topper RM**

The tricky part of _____ algorithm(s) is/are, how to detect whether the addition of an edge will create a cycle in viable set A.

A. DFS

B. Prim's

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C. Kruskal's **Page 159**

D. Both Kruskal's and Prim's

Question No:106

(Marks:1)

Vu-Topper RM

As the Kruskal's algorithm runs, the edges in viable set A induce a _____ on the vertices.

- A. Set
- B. Tree
- C. Graph
- D. Forest**

Page 147

Question No:107

(Marks:1)

Vu-Topper RM

In Dijkstra's algorithm the estimated value of source vertex $d[s]$ is:

- A. Equal to 1
- B. Equal to 0**
- C. Greater than 0
- D. Greater than 1

Page 154

Question No:108

(Marks:1)

Vu-Topper RM

Equivalence relation partitions the vertices into _____ classes of mutually reachable vertices and these are the strong components

- A. Variance
- B. Non classes
- C. Equivalence**
- D. Non equivalence

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Question No:109

(Marks:1)

Vu-Topper RM

The ancestor and descendent relation can be nicely inferred by the _____ lemma.

- A. Node
- B. Division
- C. Addition
- D. Parenthesis**

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Question No:110

(Marks:1)

Vu-Topper RM

Networks are _____ in the sense that it is possible from any location in the network to reach any other location in the digraph.

A. Not graphs.

B. Complete

Page 135

C. Incomplete

D. Transportation

Question No:111

(Marks:1)

Vu-Topper RM

In strong components algorithm, first of all DFS is run for getting _____ times of vertices.

A. Start

Page 138

B. Finish

C. Middle

D. Both start & finish

Question No:112

(Marks:1)

Vu-Topper RM

Timestamp structure of _____ is used in determining the strong components of a digraph.

A. DFS

Google

B. BFS

C. MST

D. Both DFS, BFS

Question No:113

(Marks:1)

Vu-Topper RM

In Timestamped DFS-cycles lemma, if edge (u, v) is a tree, forward or cross edge, then _____

A. $f[u] > f[v]$

B. $f[u] < f[v]$

C. $f[u] \leq f[v]$

D. $f[u] \geq f[v]$

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Question No:114

(Marks:1)

Vu-Topper RM

The component digraph is necessarily _____.

- A. Cyclic
- B. Strong
- C. Acyclic**
- D. Straight

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Question No:115

(Marks:1)

Vu-Topper RM

In computing the _____ components of a digraph, vertices of the digraph are partitioned into subsets.

- A. Best
- B. Worst
- C. Weakly connected
- D. Strongly connected**

Page 135

Question No:116

(Marks:1)

Vu-Topper RM

There exist a unique path between any _____ vertices of a free tree.

- A. One
- B. Two**
- C. five
- D. three

Question No:117

(Marks:1)

Vu-Topper RM

The _____ given by DFS allow us to determine whether the graph contains any cycles.

- A. Order
- B. Time stamps**
- C. BFS traversing
- D. Topological sort

Page 130

Question No:118

(Marks:1)

Vu-Topper RM

Forward edge is:

- A. (u, v) where u is a proper descendent of v in the tree.

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B. (u, v) where v is a proper descendent of u in the tree.

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C. (u, v) where v is a proper ancestor of u in the tree.

D. (u, v) where u is a proper ancestor of v in the tree.

Question No:119

(Marks:1)

Vu-Topper RM

Once you enter a strong component, every vertex in the component is

_____.

A. Removed

B. Reachable

Page 137

C. not reachable

D. reachable some times

Question No:120

(Marks:1)

Vu-Topper RM

In Generic approach determining of Greedy MST, we maintain a subset

A of _____.

A. Paths

B. Edges

Page 143

C. Cycles

D. Vertices

Question No:121

(Marks:1)

Vu-Topper RM

If you find yourself in maze the better traversal approach will be :

A. DFS

B. BFS

C. Level order

D. DFS

Question No:122

(Marks:1)

Vu-Topper RM

Computing the strongly connected components of a digraph is a/an

_____ of the problem to determine whether a digraph is strongly connected or not.

A. Size

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- B. Connection
- C. Optimization

D. Generalization

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Question No:123

(Marks:1)

Vu-Topper RM

A free tree with n _____ have exactly $n-1$ _____.

A. vertices,edges

Page 142

- B. vertices,nodes
- C. nodes,vertices
- D. vertices,nodes

Question No:124

(Marks:1)

Vu-Topper RM

_____ components are not affected by reversal of all edges in terms of vertices reachability.

- A. First two
- B. Last two
- C. Weakly connected

D. Strongly connected

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Question No:125

(Marks:1)

Vu-Topper RM

Adding any edge to a free tree creates a unique _____.

A. Vertex

B. Cycle

Page 142

C. Edge

D. Strong component

Question No:126

(Marks:1)

Vu-Topper RM

In computing the strongly connected components of a digraph, vertices of the digraph are _____ into subsets.

- A. Joined
- B. Deleted
- C. Created

D. Partitioned

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Question No:127

(Marks:1)

Vu-Topper RM

For _____ graphs, there is no distinction between forward and back edges.

- A. Large
- B. Medium
- C. Directed

D. Undirected

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Question No:128

(Marks:1)

Vu-Topper RM

A topological sort of a DAG is a _____ ordering of the vertices of the DAG such that for each edge (u, v) , u appears before v in the ordering.

A. Linear

Page 134

- B. Parallel
- C. Sequence
- D. Non-Linear

Question No:129

(Marks:1)

Vu-Topper RM

By breaking any edge on a cycle created in free tree, the free _____ is restored.

A. Tree

Page 142

- B. Edge
- C. Cycle
- D. Vertex

Question No:130

(Marks:1)

Vu-Topper RM

Which technique is used in the implementation of Kruskal solution for the MST?

A. Greedy Technique

Page 142

- B. Divide-and-Conquer Technique
- C. Dynamic Programming Technique

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D. The algorithm combines more than one of the above techniques i.e. Divide-and-Conquer and Dynamic Programming

Question No:131

(Marks:1)

Vu-Topper RM

A strongly connected component only apply to:

A. Directed Graph **Page 135**

B. Undirected Graph

C. Breadth First Search

D. Minimum Spanning Tree

Question No:132

(Marks:1)

Vu-Topper RM

The relationship between number of back edges and number of cycles in DFS is,

A. Both are equal

B. Back edges are half of cycles

C. Back edges are one quarter of cycles

D. There is no relationship between no. of edges and cycles

Page 131

Question No:133

(Marks:1)

Vu-Topper RM

Digraphs _____ in communication and transportation networks.

A. are used **Page 135**

B. are not used

C. parts are used

D. final value is used

Question No:134

(Marks:1)

Vu-Topper RM

Back edge is:

A. (u, v) where v is an ancestor of u in the tree. **Page 128**

B. (u,v) where u is an ancestor of v in the tree.

C. (u, v) where v is an predecessor of u in the tree.

D. None of above

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Question No:135

(Marks:1)

Vu-Topper RM

There are no _____ edges in undirected graph.

- A. Back
- B. Both
- C. Cross**
- D. Forward

Page 130

Question No:136

(Marks:1)

Vu-Topper RM

In strong components algorithm, the form of graph is used in which all the _____ of original graph G have been reversed in direction.

- A. Edges**
- B. Trees
- C. Vertices
- D. Both edges & vertices

Page 138

Question No:137

(Marks:1)

Vu-Topper RM

Which activity creates a unique cycle in a free tree:

- A. adding root
- B. adding any edge**
- C. adding any vertex
- D. adding any sub tree

Page 142

Question No:138

(Marks:1)

Vu-Topper RM

You have an adjacency list for G, what is the time complexity to compute Graph transpose G^T ?

- A. $(V + E)$**
- B. $(V E)$
- C. (V)
- D. (V^2)

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Question No:139

(Marks:1)

Vu-Topper RM

The time complexity to compute Graph transpose G^T is $(V+E)$, if you have _____ for G.

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- A. Stack
- B. Array list
- C. Complete list
- D. An adjacency list**

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Question No:140

(Marks:1)

Vu-Topper RM

In undirected graph, by convention all the edges are called _____ edges.

- A. Back**
- B. Cross
- C. Forward
- D. Both forward and back

Page 130

Question No:141

(Marks:1)

Vu-Topper RM

The _____ given by DFS allow us to determine a number of things about a graph or digraph.

- A. line stamps
- B. node stamps
- C. color stamps
- D. time stamps**

Page 130

Question No:142

(Marks:1)

Vu-Topper RM

If a subset of edges A is viable for building MST, it can not contain a/an _____.

- A. Edge
- B. Cycle**
- C. Graph
- D. Vertex

Page 143

Question No:143

(Marks:1)

Vu-Topper RM

A free tree with n vertices have exactly _____ edges.

- A. 1
- B. N

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C. n-1

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D. n+1

Question No:144

(Marks:1)

Vu-Topper RM

_____ technique is look like propagating wave-front outward.

A. Generic Traversal

B. Depth First Traversal

C. Time Stamp Traversal

D. Breath first traversal

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Question No:145

(Marks:1)

Vu-Topper RM

In Timestamped DFS, If there is a back edge (u, v) then v is an ancestor of u and by following tree edge from v to u, we get _____.

A. a line

B. a cycle

Page 131

C. a graph

D. nothing

Question No:146

(Marks:1)

Vu-Topper RM

In strong components algorithm, vertices are sorted in _____ order of finish times.

A. Any

B. Strong

C. Increasing

D. Decreasing

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Question No:147

(Marks:1)

Vu-Topper RM

A fully connected undirected graph of 5 nodes will have _____ edges.

A. 4

B. 5

C. 10

D. 15

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Question No:148

(Marks:1)

Vu-Topper RM

A digraph is strongly connected under what condition?

A. A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v .

B. A digraph is strongly connected if for every pair of vertices $u, v \in V$, u can reach v and vice versa.

C. A digraph is strongly connected if for at least one pair of vertex $u, v \in V$, u can reach v and vice versa.

D. A digraph is strongly connected if at least one third pair of vertices $u, v \in V$, u can reach v and vice versa.

Question No:149

(Marks:1)

Vu-Topper RM

We say that two vertices u and v are mutually _____ if u can reach v and vice versa.

A. Crossed

B. Forward

C. Reachable

Page 135

D. Not Reachable

Question No:150

(Marks:1)

Vu-Topper RM

In Timestamped DFS, No back edges means _____.

A. DFS

B. BFS

C. 1 cycle

D. no cycles

Page 131

Question No:151

(Marks:1)

Vu-Topper RM

Cross edge is :

A. (u, v) where u and v are not ancestor of one another

B. (u, v) where u is ancestor of v and v is not descendent of u .

C. (u, v) where u and v are not ancestor or descendent of one another

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D. (u, v) where u and v are either ancestor or descendent of one another.

Question No:152

(Marks:1)

Vu-Topper RM

In digraph $G=(V,E)$; G has cycle if and only if

A. The DFS forest has forward edge.

B. The DFS forest has back edge

Page 131

C. The DFS forest has both back and forward edge

D. BFS forest has forward edge

Question No:153

(Marks:1)

Vu-Topper RM

For each vertex $u \in (V - S)$,we associate a key _____.

A. $\text{key}[v-s]$

B. $\text{key}[s]$

C. $\text{key}[u]$

D. $\text{key}[v]$

Question No:154

(Marks:1)

Vu-Topper RM

In Kruskal's algorithm, the vertices will be stored in _____.

A. loops

B. sets

C. nodes

D. links

Question No:155

(Marks:1)

Vu-Topper RM

An edge (u, v) _____ $E - A$ is safe if A _____ $\{(u, v)\}$ is viable.

A. ϵ, U

B. U, ϵ

C. ϵ, Π

D. Π, U

Question No:156

(Marks:1)

Vu-Topper RM

Which of the following statement is true.

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- A. Kruskal's algorithm is used to find minimum spanning tree of a graph, time complexity of this algorithm is $O(EV)$
- B. Kruskal algorithm is multiple source technique for finding MST.
- C. Both I and II
- D. Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best Tree edge) when the graph has relatively few edges**

In _____ involves breaking up the problem into sub problems whose solutions can be combined to solve the global problem.

- A. Dynamic programming solution
- B. Divide and Conquer Strategy**
- C. Complexity Theory
- D. Greedy Algorithms

If matrix A of dimension 2×4 is multiplied with matrix B of dimension 4×3 , then the dimension of resultant matrix will be _____.

- A. 3×4
- B. 2×3**
- C. 2×4
- D. 4×3

The coin change problem is to find the minimum number of coins required to get the sum S of infinite coins of denominations $a_1, a_2, a_3, \dots, a_n$. This problem can be solved using _____.

- A. Divide and conquer
- B. Greedy algorithm
- C. Dynamic programming**
- D. Backtracking

If matrix A of dimension $p \times q$ is multiply with matrix B of dimension $q \times r$, then the dimension of resultant matrix is:

- A. $p \times q$

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B. $p \times r$

C. $q \times r$

D. $r \times p$

In Dynamic Programming based solution of knapsack problem, to compute entries of 'V', we will imply a(n) _____ approach.

A. Brute Force

B. Combination

C. Subjective

D. Inductive

Suppose we have 4 matrices A, B, C, D. What is correct expansion of $m[1,2]$ in chain matrix multiplication?

A. $m[1, 2] = m[1, 2] + m[2, 2] + p_0 \cdot p_1 \cdot p_2$

B. $m[1, 2] = m[1, 1] + m[2, 2] + p_0 \cdot p_1 \cdot p_3$

C. $m[1, 2] = m[1, 1] + m[2, 2] + p_0 \cdot p_1 \cdot p_2$

D. $m[1, 2] = m[1, 1] + m[1, 2] + p_0 \cdot p_1 \cdot p_2$

There are _____ entries in the Edit Distance Matrix.

A. $\Theta(n+2)$

B. $\Theta(n + 100)$

C. $\Theta(n^2)$

D. $\Theta(n)$

If there are $\Theta(n^2)$ entries in edit distance matrix then each entry $E(i, j)$ takes _____ time to compute.

A. $\Theta(n)$

B. $\Theta(n \log n)$

C. $\Theta(n^2)$

D. $\Theta(1)$

_____ algorithm based on Dynamic Programming strategy.

A. Heap Sort

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- B. Binary Tree
- C. Quick Sort
- D. Edit distance**

In Activity selection problem, intuitively _____.

- A. There are always short activities as input
- B. Short activities are not attractive
- C. We do not like long activities
- D. Duration of the activities does not matter**

In the context of encoding/decoding, fixed-length codes may not be efficient from the perspective of _____ the total quantity of data.

- A. Minimizing**
- B. Maximizing
- C. Summing
- D. Averaging

The prefix code generated by Huffman algorithm _____ the expected length of the encoded string.

- A. Maximizes
- B. Balances
- C. Keeps Constant
- D. Minimizes**

Chain matrix multiplication problem can be solved through _____ strategy.

- A. dynamic programming**
- B. greedy
- C. Implementing arrays
- D. divide and conquer

When matrix A of 5×3 is multiplied with matrix B of 3×4 then the number of multiplications required will be _____.

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- A. 12
- B. 36
- C. 60**
- D. 15

The Dynamic Programming Edit Distance Algorithm takes a total running time of _____.

- A. $\Theta(n)$
- B. $\Theta(n^2)$**
- C. $\Theta(n \log n)$
- D. $\Theta(n^3)$

If we implement the bag by using a queue, we have _____.

- A. BFS**
- B. DFS
- C. Loop
- D. Graph

Question No:0_1

(Marks:1)

Vu-Topper RM

If we encode and compress text using ASCII standard each character is represented by

- A. Fixed length code word of 4 bits
- B. Variable length code word up to 4 bits
- C. Fixed length code word of 8 bits**
- D. Variable length code word up to 8 bits

Question No:157

(Marks:1)

Vu-Topper RM

In Activity Selection, we say that two activities are non-interfering if their start-finish interval _____ overlap.

- A. Do
- B. Do not.**
- C. Sometimes

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D. Once

Question No:158

(Marks:1)

Vu-Topper RM

How many steps are involved to design the dynamic programming strategy?

A. 2

B. 3

C. 1

D. 4

ok

Question No:159

(Marks:1)

Vu-Topper RM

Bag is a _____

A. type of algorithm

ok

B. data structure

C. program

D. compiler

Question No:160

(Marks:1)

Vu-Topper RM

If a problem is in NP-complete, it must also be in NP.

True

False

Question No:161

(Marks:1)

Vu-Topper RM

The Huffman algorithm finds a optimal solution.

True

False

Question No:162

(Marks:1)

Vu-Topper RM

The Huffman algorithm finds an exponential solution

True

False

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Question No:163

(Marks:1)

Vu-Topper RM

The Huffman algorithm finds a polynomial solution

True

False

Question No:164

(Marks:1)

Vu-Topper RM

The greedy part of the Huffman encoding algorithm is to first find two nodes with smallest frequency.

True

False

Question No:165

(Marks:1)

Vu-Topper RM

The code word assigned to characters by the Huffman algorithm have the property that no code word is the prefix of any other.

True

False

Question No:166

(Marks:1)

Vu-Topper RM

Dijkstra's single source shortest path algorithm works if all edges weights are non-negative and there are no negative cost cycles.

True

False

Question No:167

(Marks:1)

Vu-Topper RM

The term "coloring" came from the original application which was in architectural design.

True

False

Question No:168

(Marks:1)

Vu-Topper RM

In the clique cover problem, for two vertices to be in the same group, they must be adjacent to each other.

True

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False

Question No:169

(Marks:1)

Vu-Topper RM

Dijkstra's algorithm is operates by maintaining a subset of vertices

True

False

Question No:170

(Marks:1)

Vu-Topper RM

We do sorting to,

- A. keep elements in random positions
- B. keep the algorithm run in linear order
- C. keep the algorithm run in $(\log n)$ order
- D. keep elements in increasing or decreasing order**

Question No:171

(Marks:1)

Vu-Topper RM

After partitioning array in Quick sort, pivot is placed in a position such that

- A. Values smaller than pivot are on left and larger than pivot are on right**
- B. Values larger than pivot are on left and smaller than pivot are on right
- C. Pivot is the first element of array
- D. Pivot is the last element of array

Question No:172

(Marks:1)

Vu-Topper RM

Merge sort is stable sort, but not an in-place algorithm

True

False

Question No:173

(Marks:1)

Vu-Topper RM

A $p \times q$ matrix A can be multiplied with a $q \times r$ matrix B. The result will be a $p \times r$ matrix C. There are $(p \cdot r)$ total entries in C and each

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takes_ to compute.

- A. O (q)**
- B. O (1)
- C. O (n^2)
- D. O (n^3)

Question No:174

(Marks:1)

Vu-Topper RM

One of the clever aspects of heaps is that they can be stored in arrays without using any -----.

- A. Pointers**
- B. constants
- C. variables
- D. functions

Question No:175

(Marks:1)

Vu-Topper RM

Merge sort requires extra array storage,

True
False

Question No:176

(Marks:1)

Vu-Topper RM

The Huffman codes provide a method of encoding data inefficiently when coded using ASCII standard.

True
False

Question No:177

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with_bits.

- A. 80**
- B. 160
- C. 320
- D. 100

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Question No:178

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 160 bits.

True

False

Question No:179

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

True

False

Question No:180

(Marks:1)

Vu-Topper RM

The greedy part of the Huffman encoding algorithm is to first find two nodes with character frequency

True

False

Question No:181

(Marks:1)

Vu-Topper RM

Huffman algorithm uses a greedy approach to generate an prefix code T that minimizes the expected length $B(T)$ of the encoded string.

True

False

Question No:182

(Marks:1)

Vu-Topper RM

An optimization problem is one in which you want to find the _____ solution.

A. Not a solution

B. An algorithm

C. Good solution

D. The best solution

ok

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Question No:183

(Marks:1)

Vu-Topper RM

Although it requires more complicated data structures, Prim's algorithm for a minimum spanning tree is better than Kruskal's when the graph has a large number of vert

True

False

Question No:184

(Marks:1)

Vu-Topper RM

If a problem is in NP, it must also be in P.

A. True

B. False

C. unknown

Question No:185

(Marks:1)

Vu-Topper RM

What is generally true of Adjacency List and Adjacency Matrix representations of graphs?

A. Lists require less space than matrices but take longer to find the weight of an edge (v1,v2)

B. Lists require less space than matrices and they are faster to find the weight of an edge (v1,v2)

C. Lists require more space than matrices and they take longer to find the weight of an edge (v1,v2)

D. Lists require more space than matrices but are faster to find the weight of an edge (v1,v2)

Question No:186

(Marks:1)

Vu-Topper RM

If a graph has v vertices and e edges then to obtain a spanning tree we have to delete

A. v edges.

B. $v - e + 5$ edges

C. $v + e$ edges.

D. None of these

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Question No:187

(Marks:1)

Vu-Topper RM

Maximum number of vertices in a Directed Graph may be $|V^2|$

True

False

Question No:188

(Marks:1)

Vu-Topper RM

The Huffman algorithm finds a (n) __solution.

A. Optimal

B. Non-optimal

C. Exponential

D. Polynomial

Question No:189

(Marks:1)

Vu-Topper RM

Edge (u, v) is a forward edge if

A. u is a proper descendant of v in the tree

B. v is a proper descendant of u in the tree pg#129

C. None of these

Question No:190

(Marks:1)

Vu-Topper RM

In counting sort, once we know the ranks, we simply _numbers to their final positions in an output array.

A. Delete

B. copy

C. Mark

D. arrange

Question No:191

(Marks:1)

Vu-Topper RM

Dynamic programming algorithms need to store the results of intermediate sub-problems.

True

False

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Question No:192

(Marks:1)

Vu-Topper RM

_____ is a graphical representation of an algorithm

- A. notation
- B. notation
- C. Flowchart**
- D. Asymptotic notation

Question No:193

(Marks:1)

Vu-Topper RM

Which of the following is calculated with **big o notation**?

- A. Lower bounds
- B. Upper bounds**
- C. Both upper and lower bound
- D. Medium bounds

Question No:194

(Marks:1)

Vu-Topper RM

Merge sort makes two recursive calls. Which statement is true after these recursive calls finish, but before the merge step?

- A. The array elements form a heap
- B. Elements in each half of the array are sorted amongst themselves.**
- C. Elements in the first half of the array are less than or equal to elements in the second half of the array
- D. None of the above

Question No:195

(Marks:1)

Vu-Topper RM

What is the solution to the recurrence $T(n) = T(n/2) + n$, $T(1) = 1$

- A. $O(\log n)$
- B. $O(n)$**
- C. $O(n \log n)$
- D. $O(2n)$

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Question No:196

(Marks:1)

Vu-Topper RM

Consider the following Huffman Tree The binary code for the string TEA is

A. 00 010

B. 011 00 010

C. 10 00 110

D. 11 10 110

Question No:197

(Marks:1)

Vu-Topper RM

A greedy algorithm does not work in phases.

True

False

Question No:198

(Marks:1)

Vu-Topper RM

Can an adjacency matrix for a directed graph ever not be square in shape?

Yes

No

Question No:199

(Marks:1)

Vu-Topper RM

One of the clever aspects of heaps is that they can be stored in arrays without using any_____.

A. Pointers

B. constants

C. variables

D. functions

Question No:200

(Marks:1)

Vu-Topper RM

Non-optimal or greedy algorithm for money change takes__

A. $O(k)$

B. $O(kN)$

C. $O(2k)$

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D. O(N)

Question No:201

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 320 bits.

True

False

Question No:202

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 100 bits.

True

False

Question No:203

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 32 bytes

True

False

Question No:204

(Marks:1)

Vu-Topper RM

Huffman algorithm uses a greedy approach to generate an **antefix** code T that minimizes the expected length B (T) of the encoded string.

True

False

Question No:205

(Marks:1)

Vu-Topper RM

Depth first search is shortest path algorithm that works on un-weighted graphs.

True

False

Question No:206

(Marks:1)

Vu-Topper RM

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Floyd-Warshall algorithm is a dynamic programming algorithm; the genius of the algorithm is in the clever recursive formulation of the shortest path problem.

True

False

Question No:207

(Marks:1)

Vu-Topper RM

Floyd-Warshall algorithm, as in the case with DP algorithms, we avoid recursive evaluation by generating a table for

A. k

B. d^k

C. True

D. False

Question No:208

(Marks:1)

Vu-Topper RM

The term coloring came from the original application which was in map drawing.

True

False

Question No:209

(Marks:1)

Vu-Topper RM

In the clique cover problem, for two vertices to be in the same group, they must be _____ each other.

A. Apart from

B. Far from

C. Near to

D. Adjacent to

Question No:210

(Marks:1)

Vu-Topper RM

Fixed-length codes may not be efficient from the perspective of the total quantity of data.

A. Minimizing

B. Averaging

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- C. Maximizing
- D. Summing

Question No:211 (Marks:1) **Vu-Topper RM**

In greedy algorithm, at each phase, you take the _____ you can get right now, without regard for future consequences.

- A. Worst
- B. Minimum
- C. Good
- D. Best**

ok

Question No:212 (Marks:1) **Vu-Topper RM**

The difference between Prim's algorithm and Dijkstra's algorithm is that Dijkstra's algorithm uses a same key.

True

False

Question No:213 (Marks:1) **Vu-Topper RM**

If there are n items, there are _____ possible combinations of the items.

- A. 2
- B. N
- C. 2^n**
- D. 3^n

Question No:214 (Marks:1) **Vu-Topper RM**

In Knapsack Problem, the thief's goal is to put items in the bag such that the _____ of the items does not exceed the limit of the bag.

- A. Value**
- B. Weight
- C. Length
- D. Balance

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Question No:215

(Marks:1)

Vu-Topper RM

The knapsack problem does not belong to the domain of optimization problems.

True

False

Question No:216

(Marks:1)

Vu-Topper RM

In Huffman encoding, for a given message string, the frequency of occurrence (relative probability) of each character in the message is determined last.

True

False

Question No:217

(Marks:1)

Vu-Topper RM

Fixed-length codes are known for easy break up of a string into its individual characters.

True

False

Question No:218

(Marks:1)

Vu-Topper RM

In Knapsack Problem, value and weight both are to be under consideration.

True

False

Question No:219

(Marks:1)

Vu-Topper RM

Multiplication is ____.

A. $\log n$

B. n

C. n^2

D. n^3

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Question No:220

(Marks:1)

Vu-Topper RM

In DP based solution of knapsack problem, to compute entries of V we will imply a/an___approach.

A. Subjective

B. Inductive

C. Brute force

D. Combination

Question No:221

(Marks:1)

Vu-Topper RM

A greedy algorithm sometimes works well for optimization problems.
True

False

Question No:222

(Marks:1)

Vu-Topper RM

In Huffman encoding, frequency of each character can be determined by parsing the message and how many times each character (or symbol) appears.

A. Printing

B. Incrementing

C. Counting

D. Deleting

Question No:223

(Marks:1)

Vu-Topper RM

Greedy algorithm can do very poorly for some problems.

True

False

Question No:224

(Marks:1)

Vu-Topper RM

The Huffman codes provide a method of___data efficiently.

A. Reading

B. Encoding

C. Decoding

D. Printing

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Question No:225

(Marks:1)

Vu-Topper RM

In__based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

A. Brute force

B. Dynamic programming

Question No:226

(Marks:1)

Vu-Topper RM

In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

True

False

Question No:227

(Marks:1)

Vu-Topper RM

_____problem, we want to find the best solution.

A. Minimization

B. Averaging

C. Optimization

D. Maximization

Question No:228

(Marks:1)

Vu-Topper RM

Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

True

False

Question No:229

(Marks:1)

Vu-Topper RM

Huffman algorithm generates an optimum prefix code.

True

False

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Question No:230

(Marks:1)

Vu-Topper RM

If the string "lmncde" is coded with ASCII code, the message length would be _____ bits.

A. 24

B. 36

C. 48

6*8=48

ok

D. 60

Question No:231

(Marks:1)

Vu-Topper RM

There are _____ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

A. 2

B. 3

C. 4

Question No:232

(Marks:1)

Vu-Topper RM

A number of lectures are to be given in a single lecture hall. Optimum scheduling for this is an example of Activity selection.

True

False

Question No:233

(Marks:1)

Vu-Topper RM

The activity scheduling is a simple scheduling problem for which the greedy algorithm approach provides a/an _____ solution.

A. Simple

B. Sub optimal

C. Optimal

ok

D. Non optimal

Question No:234

(Marks:1)

Vu-Topper RM

The string |xyz|, if coded with ASCII code, the message length would be 24 bits.

True

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False

Question No:235

(Marks:1)

Vu-Topper RM

An application problem is one in which you want to find, not just a solution, but the ____ solution.

A. Simple

B. Good

C. Best

Question No:236

(Marks:1)

Vu-Topper RM

Suppose that a graph $G = (V, E)$ is implemented using adjacency lists.

What is the complexity of a breadth-first traversal of G ?

A. $O(|V|^2)$

B. $O(|V| + |E|)$

C. $O(|V|^2|E|)$

D. $O(|V| + |E|)$

Question No:237

(Marks:1)

Vu-Topper RM

Which is true statement?

A. Breadth first search is shortest path algorithm that works on un-weighted graphs

B. Depth first search is shortest path algorithm that works on un-weighted graphs.

C. Both of above are true.

D. None of above are true.

Question No:238

(Marks:1)

Vu-Topper RM

Using ASCII standard the string "abacdaacacwe" will be encoded with _____ bits

A. 64

B. 128

C. 96

D. 120

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Question No:239

(Marks:1)

Vu-Topper RM

Kruskal's algorithm (choose best non-cycle edge) is better than Prim's (choose best tree edge) when the graph has relatively few edges.

True

False

Question No:240

(Marks:1)

Vu-Topper RM

What algorithm technique is used in the implementation of Kruskal solution for the MST?

A. Greedy Technique

B. Divide-and-Conquer Technique

C. Dynamic Programming Technique

D. The algorithm combines more than one of the above techniques.

Question No:241

(Marks:1)

Vu-Topper RM

There is relationship between number of back edges and number of cycles in DFS

A. Both are equal.

B. Cycles are half of back edges.

C. Cycles are one fourth of back edges.

D. There is no relationship between back edges and number of cycles.

Question No:242

(Marks:1)

Vu-Topper RM

In in-place sorting algorithm is one that uses arrays for storage:

A. An additional array

B. No additional array

C. Both of above may be true according to algorithm.

D. More than 3 arrays of one dimension.

Question No:243

(Marks:1)

Vu-Topper RM

In stable sorting algorithm

A. One array is used

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B. In which duplicating elements are not handled.

C. More than one arrays are required.

D. Duplicating elements remain in same relative position after sorting.

Question No:244

(Marks:1)

Vu-Topper RM

Which sorting algorithm is faster :

A. $O(n^2)$

B. $O(n \log n)$

C. $O(n+k)$

D. $O(n^3)$

Question No:245

(Marks:1)

Vu-Topper RM

In Quick sort algorithm, constants hidden in $T(n \lg n)$ are

A. Large

B. Medium

C. Not known

D. Small

Question No:246

(Marks:1)

Vu-Topper RM

Quick sort is based on divide and conquer paradigm; we divide the problem on base of pivot element and:

A. There is explicit combine process as well to conquer the solution.

B. No work is needed to combine the sub-arrays, the array is already sorted.

C. Merging the sub arrays

D. None of the above

Question No:247

(Marks:1)

Vu-Topper RM

Which may be stable sort:

A. Bubble sort

B. Insertion sort

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- C. Selection sort
D. Both of above

Question No:248 (Marks:1) **Vu-Topper RM**

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent __ series in the analysis,

- A. linear
B. arithmetic
C. exponent
D. Geometric

Question No:249 (Marks:1) **Vu-Topper RM**

Which may be stable sort:

- A. Bubble sort
B. Insertion sort
C. Selection sort
D. Both of above

Question No:250 (Marks:1) **Vu-Topper RM**

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent __ series in the analysis,

- A. linear
B. arithmetic
C. exponent
D. Geometric

Question No:251 (Marks:1) **Vu-Topper RM**

How much time merge sort takes for an array of numbers?

- A. $T(n^2)$
B. $T(n)$
C. $T(\log n)$

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D. $T(n \log n)$

Question No:252

(Marks:1)

Vu-Topper RM

We do sorting to,

A. keep elements in random positions.

B. keep the algorithm run in linear order.

C. keep the algorithm run in $(\log n)$ order.

D. keep elements in increasing or decreasing order.

Question No:253

(Marks:1)

Vu-Topper RM

Dynamic programming algorithms need to store the results of intermediate sub-problems.

True

False

Question No:254

(Marks:1)

Vu-Topper RM

Dijkstras single source shortest path algorithm works if all edges weights are non negative and there are no negative cost cycles.

False

True

Question No:255

(Marks:1)

Vu-Topper RM

Dijkstra's single source shortest path algorithm works if all edges weights are negative and there are no negative cost cycles.

True

False

Question No:256

(Marks:1)

Vu-Topper RM

In the clique cover problem, for two vertices to be in the same group, they must be ___ each other.

A. Apart from

B. Far from

C. Near to

D. Adjacent to

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Question No:257

(Marks:1)

Vu-Topper RM

Fixed-length codes may not be efficient from the perspective of _____ the total quantity of data.

A. Minimizing

B. Averaging

C. Maximizing

D. Summing

Question No:258

(Marks:1)

Vu-Topper RM

In__based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

Brute force

Dynamic programming

Question No:259

(Marks:1)

Vu-Topper RM

In brute force based solution of knapsack problem, we consider 2 cases, Leave object Or Take object.

True

False

Question No:260

(Marks:1)

Vu-Topper RM

_____problem, we want to find the best solution.

A. Minimization

B. Averaging

C. Optimization

D. Maximization

Question No:261

(Marks:1)

Vu-Topper RM

Using ASCII standard the string abacdaacac will be encoded with 10 bytes.

True

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False

Question No:262

(Marks:1)

Vu-Topper RM

How many elements do we eliminate in each time for the Analysis of Selection algorithm?

- A. $(n / 2) + n$ elements
- B. $n / 4$ elements
- C. $2n$ elements
- D. $n / 2$ elements**

Question No:263

(Marks:1)

Vu-Topper RM

Slow sorting algorithms run in,

- A. $T(n^2)$**
- B. $T(n)$
- C. $T(\log n)$
- D. $T(n \log n)$

Question No:264

(Marks:1)

Vu-Topper RM

Counting sort is suitable to sort the elements in range 1 to k:

- A. K is large
- B. K is small**
- C. K may be large or small
- D. None

Question No:265

(Marks:1)

Vu-Topper RM

Heaps can be stored in arrays without using any pointers; this is due to the __nature of the binary tree,

- A. left-complete**
- B. right-complete
- C. tree nodes
- D. tree leaves

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Question No:266

(Marks:1)

Vu-Topper RM

A heap is a left-complete binary tree that conforms to the __

A. increasing order only

B. heap order

C. decreasing order only

D. (log n) order

Question No:267

(Marks:1)

Vu-Topper RM

Divide-and-conquer as breaking the problem into a small number of

A. pivot

B. Sieve

C. smaller sub problems

D. Selection

Question No:268

(Marks:1)

Vu-Topper RM

In Sieve Technique we do not know which item is of interest

True

False

Question No:269

(Marks:1)

Vu-Topper RM

The recurrence relation of Tower of Hanoi is given below $T(n) = \begin{cases} 1 & \text{if } n=1 \\ 2T(n-1) & \text{if } n > 1 \end{cases}$ In order to move a tower of 5 rings from one peg to another, how many ring moves are required?

A. 16

B. 10

C. 32

D. 31

Question No:270

(Marks:1)

Vu-Topper RM

For the heap sort, access to nodes involves simple ____ operations.

A. arithmetic

B. binary

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- C.algebraic
- D.logarithmic

Question No:271

(Marks:1)

Vu-Topper RM

For the sieve technique we solve the problem,

- A.recursively**
- B. mathematically
- C. precisely
- D. accurately

Question No:272

(Marks:1)

Vu-Topper RM

The sieve technique works in___as follows

- A.phases**
- B. numbers
- C.integers
- D.routines

Question No:273

(Marks:1)

Vu-Topper RM

A (an)___is a left-complete binary tree that conforms to the heap order

- A.heap**
- B. binary tree
- C. binary search tree
- D.array

Question No:274

(Marks:1)

Vu-Topper RM

The sieve technique is a special case, where the number of sub problems is just

- A. 5
- B. Many
- C.1**
- D. Few

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Question No:275

(Marks:1)

Vu-Topper RM

Analysis of Selection algorithm ends up with,

- A. $T(n)$
- B. $T(1 / 1 + n)$
- C. $T(n / 2)$
- D. $T((n / 2) + n)$**

Question No:276

(Marks:1)

Vu-Topper RM

For the heap sort we store the tree nodes in

- A. level-order traversal**
- B. in-order traversal
- C. pre-order traversal
- D. post-order traversal

Question No:277

(Marks:1)

Vu-Topper RM

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,

- A. divide-and-conquer**
- B. decrease and conquer
- C. greedy nature
- D. 2-dimension Maxima

Question No:278

(Marks:1)

Vu-Topper RM

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____

- A. n items**
- B. phases
- C. pointers
- D. constant

Question No:279

(Marks:1)

Vu-Topper RM

Memorization is?

- A. To store previous results for future use

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B. To avoid these unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later

C. To make the process accurate

D. None of the above

Question No:280

(Marks:1)

Vu-Topper RM

Quick sort is

A. Not stable but in place

B. Stable & in place

C. Stable but not in place

D. Sometime stable & some times in place

Question No:281

(Marks:1)

Vu-Topper RM

One example of in place but not stable algorithm is

A. Merger Sort

B. Quick Sort

B. Continuation Sort

C. Bubble Sort

Question No:282

(Marks:1)

Vu-Topper RM

Continuation sort is suitable to sort the elements in range 1 to k

A. K is Large

B. K is not known

C. K may be small or large

D. K is small.

Question No:283

(Marks:1)

Vu-Topper RM

Which may be a stable sort?

A. Merger

B. Insertion

C. None of the above

D. Both above

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Question No:284

(Marks:1)

Vu-Topper RM

An in place sorting algorithm is one that uses ___ arrays for storage

- A. Two dimensional arrays
- B. More than one array
- C. No Additional Array**
- D. None of the above

Question No:285

(Marks:1)

Vu-Topper RM

Continuing sort has time complexity of ?

- A. $O(n)$**
- B. $O(n+k)$
- C. $O(n \log n)$
- D. $O(k)$

Question No:286

(Marks:1)

Vu-Topper RM

single item from a larger set of _

- A. phases
- B. n items**
- C. pointers
- D. constant

Question No:287

(Marks:1)

Vu-Topper RM

For the Sieve Technique we take time

- A. $T(nk)$**
- B. $T(n / 3)$
- C. n^2
- D. $n/3$

Question No:288

(Marks:1)

Vu-Topper RM

Due to left complete nature of binary tree, the heap can be stored in

- A. Arrays**
- B. Structures

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- C. Link List
- D. Stack

Question No:289 (Marks:1) **Vu-Topper RM**

What type of instructions Random Access Machine (RAM) can execute?

- A. Algebraic and logic
- B. Geometric and arithmetic
- C. Arithmetic and logic**
- D. Parallel and recursive

Question No:290 (Marks:1) **Vu-Topper RM**

What is the total time to heapify?

- A. $O(\log n)$**
- B. $O(n \log n)$
- C. $O(n^2 \log n)$
- D. $O(\log^2 n)$

Question No:291 (Marks:1) **Vu-Topper RM**

word Algorithm comes from the name of the muslim author _____

Abu Ja'far Mohammad ibn Musa al-Khowarizmi.

Question No:292 (Marks:1) **Vu-Topper RM Al-**

Khwarizmi's work was written in a book titled _

al Kitab al-mukhtasar fi hisab al-jabr wa'l-muqabalah

Question No:293 (Marks:1) **Vu-Topper RM**

Random access machine or RAM is a/an

- A. Machine build by Al-Khwarizmi
- B. Mechanical machine
- C. Electronics machine
- D. Mathematical model**

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Question No:294

(Marks:1)

Vu-Topper RM A

RAM is an idealized machine with____ random-access memory.

- A. 256MB
- B. 512MB
- C. 100GB
- D. an infinitely large**

Question No:295

(Marks:1)

Vu-Topper RM

What will be the total number of max comparisons if we run brute-force maxima algorithm with n elements?

- A. 2 n
- B. 2 n n
- C. n**
- D. 8 n

Question No:296

(Marks:1)

Vu-Topper RM

Consider the following code: For(j=1; j<n; j++)

For(k=1; k<15; k++)

For(l=5; l<n; l++)

```
{  
Do_something_constant();  
}
```

What is the order of execution for this code.

- A. $O(n)$**
- B. $O(n^3)$
- C. $O(n^2 \log n)$
- D. $O(n^2)$

Question No:297

(Marks:1)

Vu-Topper RM

Is it possible to sort without making comparisons?

Yes

No

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Question No:298

(Marks:1)

Vu-Topper RM

When we call heapify then at each level the comparison performed takes time

A. It will take $\Theta(1)$

B. Time will vary according to the nature of input data

C. It can not be predicted

D. It will take $\Theta(\log n)$

Question No:299

(Marks:1)

Vu-Topper RM

In Quick sort, we don't have the control over the sizes of recursive calls

A. True

B. False

C. Less information to decide

D. Either true or false

Question No:300

(Marks:1)

Vu-Topper RM

For Chain Matrix Multiplication we cannot use divide and conquer approach because,

A. We do not know the optimum k

B. We use divide and conquer for sorting only

C. We can easily perform it in linear time

D. Size of data is not given

Question No:301

(Marks:1)

Vu-Topper RM

Suppose we have three items as shown in the following table, and suppose the capacity of the knapsack is 50 i.e. $W = 50$.

Item	Value	Weight
1	60	10
2	100	20

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3	120	30
---	-----	----

The optimal solution is to pick

- A. Items 1 and 2
- B. Items 1 and 3
- C. Items 2 and 3**
- D. None of these

Question No:302

(Marks:1)

Vu-Topper RM

who invented the quick sort

C.A.R. Hoare

Question No:303

(Marks:1)

Vu-Topper RM

main elements to a divide-and-conquer

Divide, conquer, combine.

Question No:304

(Marks:1)

Vu-Topper RM

Mergesort is a stable algorithm but not an in-place algorithm.

True

False

Question No:305

(Marks:1)

Vu-Topper RM

Counting sort the numbers to be sorted are in the range 1 to k where k is small.

True

False

Question No:306

(Marks:1)

Vu-Topper RM

In selection algorithm, because we eliminate a constant fraction of the array with each phase, we get the

A. Convergent geometric series

B. Divergent geometric series

C. None of these

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Question No:307

(Marks:1)

Vu-Topper RM

In RAM model instructions are executed

One after another

Parallel

Question No:308

(Marks:1)

Vu-Topper RM

Due to left-complete nature of binary tree, heaps can be stored in

A. Link list

B. Structure

C. Array

D. None of above

Question No:309

(Marks:1)

Vu-Topper RM

The time assumed for each basic operation to execute on RAM model of computation is-----

A. Constant

B. Infinite

C. Continuous

D. Variable

Question No:310

(Marks:1)

Vu-Topper RM

If the indices passed to merge sort algorithm are not equal, the algorithm may return immediately.

True

False

Question No:311

(Marks:1)

Vu-Topper RM

Brute-force algorithm uses no intelligence in pruning out decisions.

True

False

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Question No:312

(Marks:1)

Vu-Topper RM

In analysis, the Upper Bound means the function grows asymptotically no faster than its largest term.

True

False

Question No:313

(Marks:1)

Vu-Topper RM

For small values of n, any algorithm is fast enough. Running time does become an issue when n gets large

True

False

Question No:314

(Marks:1)

Vu-Topper RM

In simple brute-force algorithm, we give no thought to efficiency.

True

False

Question No:315

(Marks:1)

Vu-Topper RM

The ancient Roman politicians understood an important principle of good algorithm design that is plan-sweep algorithm.

True

False

Question No:316

(Marks:1)

Vu-Topper RM

If the indices passed to merge sort algorithm are __, then this means that there is only one element to sort.

A. Small

B. Large

C. Equal

D. Not Equal

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Question No:317

(Marks:1)

Vu-Topper RM

In pseudo code, the level of details depends on intended audience of the algorithm.

True

False

Question No:318

(Marks:1)

Vu-Topper RM

In 2d-space a point is said to be__if it is not dominated by any other point in that space.

A. Member

B. Minimal

C. Maximal

D. Joint

Question No:319

(Marks:1)

Vu-Topper RM

An algorithm is a mathematical entity that is dependent on a specific programming language.

True

False

Question No:320

(Marks:1)

Vu-Topper RM

The running time of an algorithm would not depend upon the optimization by the compiler but that of an implementation of the algorithm would depend on it.

True

False

Question No:321

(Marks:1)

Vu-Topper RM

$F(n)$ and $g(n)$ are asymptotically equivalent. This means that they have essentially the same__ for large n .

A. Results

B. Variables

C. Growth rates

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D. Size

Question No:322

(Marks:1)

Vu-Topper RM

$8n^2 + 2n - 3$ will eventually exceed $c_2 \cdot (n)$ no matter how large we make c_2 .

True

False

Question No:323

(Marks:1)

Vu-Topper RM

If we associate (x, y) integers pair to cars where x is the speed of the car and y is the negation of the price. High y value for a car means a _____ car.

A. Fast

B. Slow

C. Cheap

D. Expensive

Question No:324

(Marks:1)

Vu-Topper RM

While solving Selection problem, in Sieve technique we partition input data

A. In increasing order

B. In decreasing order

C. According to Pivot

D. Randomly

Question No:325

(Marks:1)

Vu-Topper RM

In Sieve Technique, we know the item of interest.

True

False

Question No:326

(Marks:1)

Vu-Topper RM

In Heap Sort algorithm, we build_for ascending sort.

Max heap

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Min heap

Question No:327

(Marks:1)

Vu-Topper RM

Quick sort is best from the perspective of Locality of reference.

True

False

Question No:328

(Marks:1)

Vu-Topper RM

While Sorting, the ordered domain means for any two input elements x and y

_____ satisfies only.

A. $x < y$

B. $x > y$

C. $x = y$

D. All of the above

Question No:329

(Marks:1)

Vu-Topper RM

Algorithm is a mathematical entity, which is independent of a specific machine and operating system.

True

False

Question No:330

(Marks:1)

Vu-Topper RM

In Heap Sort algorithm, the total running time for Heapify procedure is

A. Theta ($\log n$)

B. Order ($\log n$)

C. Omega ($\log n$)

D. O (1)

Question No:331

(Marks:1)

Vu-Topper RM

A point p in 2-dimensional space is usually given by its integer coordinate(s)_____

A. p.x only p.y

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B. only p.x & p.z

C. p.x & p.y

Question No:332

(Marks:1)

Vu-Topper RM

In Heap Sort algorithm, the maximum levels an element can move upward is

A. Theta ($\log n$)

B. Order ($\log n$)

C. Omega ($\log n$)

D. O

Question No:333

(Marks:1)

Vu-Topper RM

Floor and ceiling are__to calculate while analyzing algorithms.

Very easy

Usually considered difficult.

Question No:334

(Marks:1)

Vu-Topper RM

_____is one of the few problems, where provable lower bounds exist on how fast we can sort.

A. Searching

B. Sorting

C. Both Searching & Sorting

D. Graphing

Question No:335

(Marks:1)

Vu-Topper RM

A RAM is an idealized algorithm with takes an infinitely large random-access memory.

True

False

Question No:336

(Marks:1)

Vu-Topper RM

Upper bound requires that there exist positive constants c_2 and n_0 such that $f(n) \geq c_2 n$ for all $n \leq n_0$ (ye question ghalat lag raha hai mujhae)

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- A. Less than
- B. Equal to or Less than**
- C. Equal or Greater than
- D. Greater than

Question No:337 (Marks:1) **Vu-Topper RM**

In Heap Sort algorithm, if heap property is violated ____

- A. We call Build heap procedure
- B. We call Heapify procedure**
- C. We ignore
- D. Heap property can never be violated

Question No:338 (Marks:1) **Vu-Topper RM**

In ____ we have to find rank of an element from given input.

- A. Merge sort algorithm
- B. Selection problem**
- C. Brute force technique
- D. Plane Sweep algorithm

Question No:339 (Marks:1) **Vu-Topper RM**

A point p in 2-dimensional space is usually given by its integer coordinate(s) ____

- A. p.x only
- B. p.y only
- C. p.x & p.z
- D. p.x & p.y**

Question No:340 (Marks:1) **Vu-Topper RM**

In Quick Sort Constants hidden in $T(n \log n)$ are

- A. Large
- B. Medium
- C. Small**
- D. Not Known

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Question No:341

(Marks:1)

Vu-Topper RM

The running time of quick sort depends heavily on the selection of

- A. No of inputs
- B. Arrangement of elements in array
- C. Size o elements
- D. Pivot elements**

Question No:342

(Marks:1)

Vu-Topper RM

Choose one Counting sort has timecomplexity:

- A. $O(n)$** **Page 58**
- B. $O(n+k)$
- C. $O(k)$
- D. $O(n \log n)$

Question No:343

(Marks:1)

Vu-Topper RM

In place stable sorting algorithm.

- A. If duplicate elements remain in the same relative position after sorting**
- B. One array is used
- C. More than one arrays are required
- D. Duplicating elements not handled

Question No:344

(Marks:1)

Vu-Topper RM

Cont sort is suitable to sort the elements in range 1 to k

- A. K is Large
- B. K is not known
- C. K may be small or large
- D. K is small**

Question No:345

(Marks:1)

Vu-Topper RM

The sieve technique works where we have to find_item(s) from a large input.

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A. Single

B. Two

C. Three

D. Similar

Question No:346

(Marks:1)

Vu-Topper RM

In strong components algorithm, first of all DFS is run for computing finish times of vertices.

A. $[d[u], f[u]] \subseteq [d[v], f[v]]$

B. $[d[u], f[u]] \supseteq [d[v], f[v]]$

C. unrelated

D. Disjoint

Question No:347

(Marks:1)

Vu-Topper RM

For the sieve technique we solve the problem,

A. recursively

B. mathematically

C. precisely

D. accurately

Question No:348

(Marks:1)

Vu-Topper RM

number of nodes in a complete binary tree of height h is

A. $2^{(h+1)} - 1$

B. $2 * (h+1) - 1$

C. $2 * (h+1)$

D. $((h+1)^2) - 1$

Question No:349

(Marks:1)

Vu-Topper RM

Heaps can be stored in arrays without using any pointers; this is due to the

_____ nature of the binary tree,

A. left-complete

B. right-complete

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- C. tree nodes
- D. tree leaves

Question No:350

(Marks:1)

Vu-Topper RM

Sorting is one of the few problems where provable bounds exist on how fast we can sort,

- A. upper
- B. lower**
- C. average
- D. log n

Question No:351

(Marks:1)

Vu-Topper RM

The analysis of Selection algorithm shows the total running time is indeed

_____ in no

- A. arithmetic
- B. geometric
- C. linear**
- D. orthogonal

Question No:352

(Marks:1)

Vu-Topper RM

What algorithm technique is used in the implementation of Kruskal solution for the MST?

- A. Greedy Technique**
- B. Divide-and-Conquer Technique
- C. Dynamic Programming Technique
- D. The algorithm combines more than one of the above techniques

Question No:353

(Marks:1)

Vu-Topper RM

one If you find yourself in a maze the better traversal approach will be :

- A. BFS**
- B. BFS and DFS both are valid

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- C. Level order
- D. DFS

Question No:354

(Marks:1)

Vu-Topper RM

Suppose that a graph $G = (V, E)$ is implemented using adjacency lists. What is the complexity of a breadth-first traversal of G ?

- A. $O(|V|^2)$
- B. $O(|V| + |E|)$
- C. $O(|V|^2|E|)$
- D. $O(|V| + |E|)$**

Question No:355

(Marks:1)

Vu-Topper RM

There are __ nested loops in DP based algorithm for computing the minimum cost of chain matrix multiplication.

- A. 2
- B. 3**
- C. 4
- D. 5

Question No:356

(Marks:1)

Vu-Topper RM

Counting Money problem is an example which cannot be optimally solved by greedy algorithm.

True
False

Question No:357

(Marks:1)

Vu-Topper RM

Who invented Quick sort procedure?

- A. Hoare**
- B. Sedgewick
- C. Mellroy
- D. Coreman

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Question No:358

(Marks:1)

Vu-Topper RM

An activity scheduling algorithm, the width of a rectangle --.

- A. Is always ignored
- B. Direct toward recursion
- C. Should be maximized

D. Indicates the duration of an activity

Question No:359

(Marks:1)

Vu-Topper RM

In recursive formulation of Knapsack Problem: $V[0, j] = \underline{\hspace{2cm}}$ for $j \geq 0$

A. -1

B. 0 ok

C. 1

D. 2

Question No:360

(Marks:1)

Vu-Topper RM

The probability in Huffman encoding is the number of occurrences of a character divided by the total in the message.

- A. Numbers
- B. Frequencies
- C. Strings

D. Characters ok

Question No:361

(Marks:1)

Vu-Topper RM

Graphs can be represented by an and .

A. queue , stack

B. adjacency list , adjacency matrix ok

C. adjacency right , adjacency left

D. Binary , linear

Question No:362

(Marks:1)

Vu-Topper RM

G power T for graph can be computed in

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- A. $\Theta(V E)$
- B. $\Theta(V \log E)$
- C. $\Theta(E \log V)$
- D. $\Theta(V + E)$**

Question No:363

(Marks:1)

Vu-Topper RM

Adding any edge to free tree

- A. Keep it free tree and increase the size of the tree.**
- B. Create a unique cycle.
- C. Not allow to add edge to free tree.
- D. Create multiple cycles.

Question No:364

(Marks:1)

Vu-Topper RM

Which sorting algorithm is faster :

- A. $O(n^2)$
- B. $O(n \log n)$
- C. $O(n+k)$**
- D. $O(n^3)$

Question No:365

(Marks:1)

Vu-Topper RM

Networks are complete in the sense that it is possible from any location in the network to reach any other location in the digraph.

True

False

Question No:366

(Marks:1)

Vu-Topper RM

According to parenthesis lemma vertex u is a descendent of v vertex if and only if:

$f[d[u], f[u]] \subseteq [d[v], f[v]]$ (Page 129)

Question No:367

(Marks:1)

Vu-Topper RM

Runtime complexity of Prim's algorithm is _____.

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- A. $V \log V$
- B. $E \log V$**
- C. $\log V$
- D. None of the above

Question No:368

(Marks:1)

Vu-Topper RM

Adding any edge to a free tree creates a unique cycle.

True

False

Question No:369

(Marks:1)

Vu-Topper RM

For undirected graph, there is no distinction between forward and back edges.

True

False

Question No:370

(Marks:1)

Vu-Topper RM

in strong components algorithm, first of all DFS is run for computing finish times of vertices.

True

False

Question No:371

(Marks:1)

Vu-Topper RM

Which method is preferable for dealing with chain matrix multiplication?

- A. Divide and conquer strategy
- B. Dynamic programming formulation**
- C. Graph theory
- D. Greedy Approach

Question No:372

(Marks:1)

Vu-Topper RM

Huffman algorithm produces the _____ prefix code tree.

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- A. Worst
- B. Best
- C. Optimal**
- D. Better

ok

Question No:373

(Marks:1)

Vu-Topper RM

A _____ w is adjacent to vertex v if there is an edge from v to w.

- A. Acyclic
- B. Vertex**
- C. Loop
- D. Cycle

ok

Question No:374

(Marks:1)

Vu-Topper RM

Using ASCII standard the string "greedy" will be encoded with

- A. 44 bits
- B. 120 bits
- C. 40 bits
- D. 48 bits**

ok

Question No:375

(Marks:1)

Vu-Topper RM

In Activity scheduling algorithm, each activity is represented by a _____

- A. Rectangle**
- B. Square
- C. Circle
- D. Triangle

ok

Question No:376

(Marks:1)

Vu-Topper RM

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Those problems in which Greedy finds good, but not always best is called a greedy_____.

- A. Heuristic** ok
- B. Solution
- C. Result
- D. Algorithm

Question No:377 (Marks:1) **Vu-Topper RM**

The Knapsack problem belongs to the domain of _____ problems.

- A. Searching
- B. Sorting
- C. Linear solution
- D. Optimization** ok

Question No:378 (Marks:1) **Vu-Topper RM**

The general coin change problem can be solved using_____.

- A. Recursion
- B. Greedy algorithm
- C. Dynamic programming** ok
- D. Divide and conquer

Question No:379 (Marks:1) **Vu-Topper RM**

Huffman algorithm generates an optimum _____ code.

- A. Postfix
- B. Infix
- C. None of the given options
- D. Prefix** ok

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Question No:380

(Marks:1)

Vu-Topper RM

..... ways of representing graphics

A. 1

B. 2

C. 3

D. 4

Question No:381

(Marks:1)

Vu-Topper RM

Knapsack word originates from.....language

A. German

B. English

C. French

D. Norwegian

Question No:382

(Marks:1)

Vu-Topper RM

Graphs are important model for many application problems

A. Mathematical

B. Unpredictable

C. Haphazard

D. Unsystematic

Question No:383

(Marks:1)

Vu-Topper RM

Which type of algorithm is harder to prove the correctness?

A. Dynamic

B. Greedy

ok

C. Divide and conquer

D. Brute force

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Question No:384

(Marks:1)

Vu-Topper RM

_____ items are not allowed in 0/1 Knapsack problem.

A. Fractional ok

B. 0

C. 1

D. 0/1

Question No:385

(Marks:1)

Vu-Topper RM

Matrix multiplication is a(n) _____ operation.

A. Nether commutative nor associative

B. Transitive

C. Commutative

D. Associative ok

Question No:386

(Marks:1)

Vu-Topper RM

For a Diagraph $G=(V,E)$, Sum of in-degree (v) --.

A. Not equal to sum of out-degree(v)

B. = sum of out-degree(v)

C. < sum of out-degree(v)

D. > sum of out-degree(v)

Question No:387

(Marks:1)

Vu-Topper RM

DFS or BFS yields a _____ of the graph.

A. Traversed tree

B. Spanning tree Page 125

C. Simple tree

D. Free tree

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Question No:388

(Marks:1)

Vu-Topper RM

Using ASCII code, each character is represented by a fixed-length code of _____ bits per character.

- A. 4
- B. 6
- C. 10
- D. 8**

ok

Question No:389

(Marks:1)

Vu-Topper RM

In Knapsack Problem, the goal is to put items in the knapsack such that the value of the items is _____ subject to weight limit of knapsack.

- A. Minimized
- B. Decreased
- C. Maximized**
- D. None of the above given

ok

Question No:390

(Marks:1)

Vu-Topper RM

A graph is said to be acyclic if it contains _____

- A. At least one cycle
- B. Exactly one cycle
- C. Always more than one cycle
- D. No cycles**

ok

Question No:391

(Marks:1)

Vu-Topper RM

The number of edges that come out of a vertex is called the _____ of that vertex in a digraph.

- A. Post-degree
- B. in-degree
- C. out-degree**
- D. pre-degree

ok

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Question No:392

(Marks:1)

Vu-Topper RM

What general property of the list indicates that the graph has an isolated vertex?

- A. There is Null pointer at the end of list.
- B. The Isolated vertex is not handled in list.**
- C. Only one value is entered in the list.
- D. There is at least one null list.

Question No:393

(Marks:1)

Vu-Topper RM

As the Kruskal's runs, the edges in viable set A induce a _____ on the vertices.

- A. Set
- B. Graph
- C. Tree
- D. Fore**

Question No:394

(Marks:1)

Vu-Topper RM

If Matrix-A has dimensions "3x2" and Matrix-B has dimensions "2x3", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- A. 3x2
- B. 2x3
- C. 2x2
- D. 3x3** ok

Question No:395

(Marks:1)

Vu-Topper RM

A/an _____ is one in which you want to find, not just a solution, but the best solution.

- A. Optimization problem** ok
- B. Divide and Conquer
- C. NP complete problem
- D. Best problem

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Question No:396

(Marks:1)

Vu-Topper RM

Fractional Knapsack is founded on _____ method.

A. Greedy **ok**

B. Recursive

C. Divide and Conquer

D. Dynamic programming

Question No:397

(Marks:1)

Vu-Topper RM

Find the maximum value of the items which can carry using knapsack weight capacity

=50

ITEM WEIGHT	10	20	30	70
VALUE	70	20	80	200

A. 90

B. 280

C. 200

D. 100

Question No:398

(Marks:1)

Vu-Topper RM

If the graph is represented using an adjacency matrix, then Breadth-first search takes-----time.

A. $O(E+1)$

B. $O(V^2)$

C. $O(V)$

D. $O(E)$

Question No:399

(Marks:1)

Vu-Topper RM

In inductive approach of knapsack problem, we consider 2 cases,
_____ Or _____.

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- A. Recursive, Iterative
- B. Median, Mode
- C. Leave object, Take object**
- D. Sequentially, Parallel

ok

Question No:400

(Marks:1)

Vu-Topper RM

A Greedy algorithm can NOT be used to solve all the _____ problems.

- A. Dynamic programming**
- B. Memorization programming
- C. Edit-distance programming
- D. Storing value programming

ok

Question No:401

(Marks:1)

Vu-Topper RM

In Huffman encoding, the _____ is the number of occurrences of a character divided by the total characters in the message.

- A. Counting
- B. Parsing
- C. Relative Probability**
- D. Weight

ok

Question No:402

(Marks:1)

Vu-Topper RM

The Binary Tree constructed by Huffman Encoding is a:

- A. Incomplete binary tree**
- B. Partial binary tree
- C. None of the given options
- D. Full binary tree

ok

Question No:403

(Marks:1)

Vu-Topper RM

Following is not the application of Edit Distance Problem.

- A. Speech recognition**
- B. Spelling correction

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- C. Ascending order
- D. Computational Molecular Biology

Question No:404 (Marks:1) **Vu-Topper RM**

Consider three matrices X, Y, Z of dimensions 1×2 , 2×3 , 3×4 respectively. The number of multiplications of (XY)Z is:

- A. 32** ok
- B. 26
- C. 24
- D. 16

Question No:405 (Marks:1) **Vu-Topper RM**

In-----Knapsack Problem, limitation is that an item can either be put in the bag or not. Fractional items are not allowed.

- A. 0
- B. 1
- C. 0/1**
- D. Fractional

Question No:406 (Marks:1) **Vu-Topper RM**

An in-place sorting algorithm is one that uses additional array for storage.

- A. Always
- B. Permanently
- C. Does not**
- D. Sometime

Question No:407 (Marks:1) **Vu-Topper RM**

If Matrix-A has dimensions " $p \times q$ " and Matrix-B has dimensions " $q \times r$ ", then multiplication of Matrix-A and Matrix-B will result a new Matrix-C having dimensions.

- A. $P \times q$
- B. $P \times r$**

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- C. $q \times r$
- D. $q \times p$

Question No:408

(Marks:1)

Vu-Topper RM

Counting sort is suitable to sort the elements in range 1 to K.

- A. K is large
- B. K is small**
- C. K may be large or small
- D. None

Question No:409

(Marks:1)

Vu-Topper RM

When matrix A of 5×3 is multiply with matrix B of 3×4 then the multiplication required is:

- A. 15
- B. 12
- C. 36
- D. 60**

Question No:410

(Marks:1)

Vu-Topper RM

-----is a linear time sorting algorithm.

- A. Merge sort
- B. Quick sort
- C. Bubble sort
- D. Radix sort**

Question No:411

(Marks:1)

Vu-Topper RM

In Dynamic Programming approach, we do not store the solution to each sub problem in case if it reappears.

True

False

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Question No:412

(Marks:1)

Vu-Topper RM

Dynamic Programming approach is usually useful in solving optimization problem.

True

False

Question No:413

(Marks:1)

Vu-Topper RM

Which of the following algorithms provides an optimal solution for the activity selection problem?

A. Divide and Conquer

B. Brute force

C. Greedy

ok

D. Recursive

Question No:414

(Marks:1)

Vu-Topper RM

A graph is _____ if every vertex can reach every other vertex.

A. Connected

ok

B. Cycle

C. Acyclic

D. Loop

Question No:415

(Marks:1)

Vu-Topper RM

In Huffman encoding when a new node is created by combining two nodes, the new node is placed in the _____.

A. Priority queue

ok

B. Linked list

C. Min heap tree

D. Graph traversal

Question No:416

(Marks:1)

Vu-Topper RM

In _____ algorithm, you hope that by choosing a local optimum at each step, you will end up at a global optimum.

A. Simple

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B. Divide and conquer

C. Greedy

ok

D. Brute Force

Question No:417

(Marks:1)

Vu-Topper RM

The string "Imncde" is coded with ASCII code, the message length would be _bits.

A. 24

B. 36

C. 48

D. 60

Question No:418

(Marks:1)

Vu-Topper RM

For graph traversal, Breadth-first search strategy _____

A. Is always recursive

B. Cannot be recursive

C. Cannot be non-recursive

D. Can be both recursive and non-recursive

ok

Question No:419

(Marks:1)

Vu-Topper RM

In activity scheduling algorithm, the width of a rectangle __

A. Is always ignored

B. Directs towards recursion

C. Should be maximized

D. Indicates the duration of an activity

Question No:420

(Marks:1)

Vu-Topper RM

If the graph is represented using an adjacency list, then Breadth-first search take time

A. $O(V^2)$

B. $O(V)$

C. $O(V+E)$

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D.O(E+1)

Question No:421

(Marks:1)

Vu-Topper RM

Suppose you are given infinite coins of 1, 2, 3, and 4. Select the ways of the minimum number of coins that required to achieve a sum of 6:

A. 1

B. 2 ok

C. 3

D. 4

Question No:422

(Marks:1)

Vu-Topper RM

Using ASCII standard the string "greedy" will be encoded with

A. 48 bitS

B. 20 bits

C. 44 bits

D. 40 bits

Question No:423

(Marks:1)

Vu-Topper RM

The Huffman codes provide a method of _____ data efficiently.

A. Reading/Writing

B. Encoding/Decoding ok

B. Divide/Conquer

C. Inserting/Deleting

Question No:424

(Marks:1)

Vu-Topper RM

In the context of activity selection algorithm, time is dominated by sorting of the activities by-----.

A. Start Times

B. Finish Times

C. Average Times

D. CPU Burst Times

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Question No:425

(Marks:1)

Vu-Topper RM

Time complexity of the “0-1” knapsack algorithm depends on _____

- A. Number of items
- B. Capacity of the knapsack
- C. Size of the Table

D. Number of items and capacity of knapsack

ok

Question No:426

(Marks:1)

Vu-Topper RM

The greedy approach gives us an optimal solution when the coins are all powers of a _____ denomination.

- A. Fixed**
- B. Variable
- C. Constant
- D. Static

ok

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